

Adaboost ML Algorithm

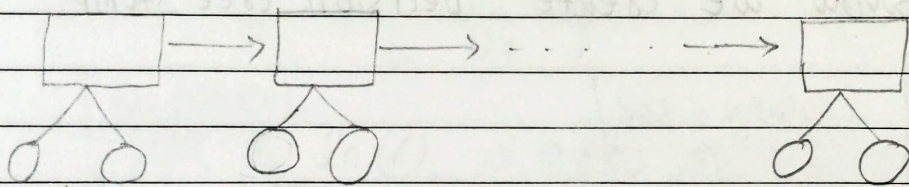
In adaboost Algorithm we assign the weights to the weak learners so that they can make Prediction well.

$$f = \alpha_1(m_1) + \alpha_2(m_2) + \dots + \alpha_n(m_n)$$

$$\text{weights} = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$$

$$\text{Decision tree stumps} = \{m_1, m_2, \dots, m_n\}$$

Stump $\Rightarrow$ It is a series connection of multiple Decision tree whose Depth is only 1.



$\Downarrow$
{weak Learner}

$\Downarrow$
Under fitting

$\begin{cases} \rightarrow \text{Train acc} \downarrow \\ \rightarrow \text{Test acc} \uparrow \end{cases} \Rightarrow \begin{cases} \text{High bias} \\ \text{Low variance} \end{cases}$

$\Downarrow$
Connection to many stumps.

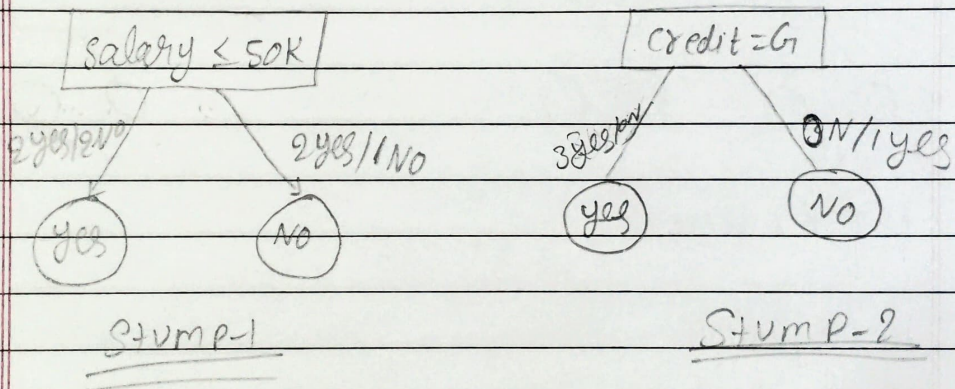
$\Downarrow$
 $\begin{cases} \text{Low bias} \\ \text{High variance} \end{cases}$

Adaboost Classifier math

Salary	Credit	Approval	Sample weight
$\leq 50K$	B	No	1/7
$\leq 50K$	G	yes	1/7
$\leq 50K$	G	yes	1/7
$> 50K$	B	No	1/7
$> 50K$	G	yes	1/7
$> 50K$	N	yes	1/7
$\leq 50K$	N	No	1/7

error

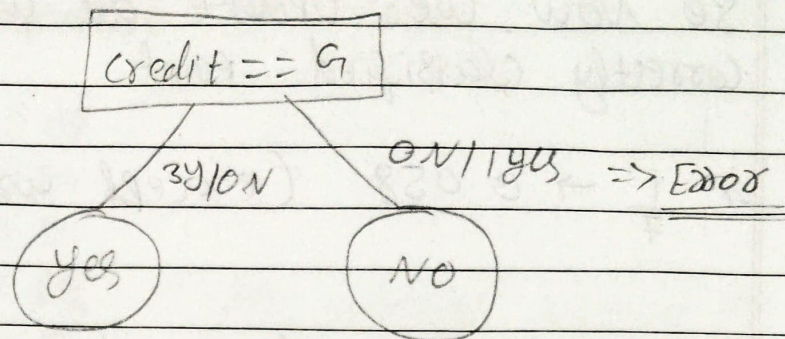
① $\Rightarrow$ Now we create decision tree stump.



After creating the Stumps or weak learners we use the entropy or Gini Index to find out which weak learner is best for starting the calculation.

② Sum of the total Errors And Performance of Stump.

we assign a sample weights to all the samples is equal.



$$\text{Total error} = \frac{1}{7}$$

$$\text{Performance of stump} = \frac{1}{2} \ln \left[\frac{1 - \text{Total error}}{\text{Total error}} \right]$$

$$\Rightarrow \frac{1}{2} \ln \left[\frac{1 - 1/7}{1/7} \right]$$

$$\Rightarrow \frac{1}{2} \ln(6) \Rightarrow 0.896 \Rightarrow \underline{\underline{\alpha \text{ (weight)}}}$$

③ Update the weights for correctly and incorrectly classified points.

$\Rightarrow$ For correct classified points $\rightarrow$ we have to decrease the weight.

$$\# \boxed{(\text{weight})_{\text{new}} = (\text{weight})_{\text{old}} \times e^{-(\text{performance})}}$$

$$w = \frac{1}{7} \times e^{-(0.896)} \Rightarrow \underline{\underline{0.058}}$$

So now we update all weights of correctly classified point.

$$\Rightarrow \frac{1}{7} \rightarrow 0.058 \text{ (except wrong one)}$$

② For Incorrect classified points:

$$\# \boxed{(\text{Weight})_{\text{new}} = (\text{weight})_{\text{old}} \times e^{(\text{performance})}}$$

$$\begin{aligned} w_{\text{new}} &= \frac{1}{7} \times e^{0.896} \\ &= \underline{0.349} \end{aligned}$$

now the updated weights for new stump is

$$\Rightarrow [0.058, 0.058, 0.058, 0.058, 0.349, 0.058, 0.058]$$

④ Normalized weight and assigning bins.

first sum of all the updated weights.

$$\Rightarrow 0.058 \times 6 + 0.349 = \underline{0.697}$$

now for finding the Normalized weight divide all weights by 0.697.

Your new normalized weight will be $\Rightarrow$

$$\Rightarrow [0.08, 0.08, 0.08, 0.08, 0.50, 0.08, 0.08] \approx 1$$

Bins assignment $\rightarrow$ a range.

0.0 - 0.08

0.08 - 0.16

0.16 - 0.24

0.24 - 0.32

0.32 - 0.40

0.40 - 0.48

$\rightarrow$ { for wrong record the bin size is more }

⑤ Select the data points for new stump.

Salary (K)	Credit	weights	N.weights	Bins
≤ 50	B	0.058	0.08	0.0 - 0.08
≤ 50	G	0.058	0.08	0.08 - 0.16
≤ 50	G	0.058	0.08	0.16 - 0.24
> 50	B	0.058	0.08	0.24 - 0.32
> 50	G	0.349	0.50	0.32 - 0.40
> 50	N	0.058	0.08	0.40 - 0.48
≤ 50	N	0.058	0.08	0.48 - 0.56
		≈ 0.697	≈ 1.0	

$\Rightarrow$ Iteration process selecting random variable from 0 - 1

S(K)	C	Approval	Random
> 50	N	yes	0.50
≤ 50	G	yes	0.10
> 50	N	yes	0.60
> 50	N	yes	0.75
> 50	B	NO	0.24
> 50	N	yes	0.42

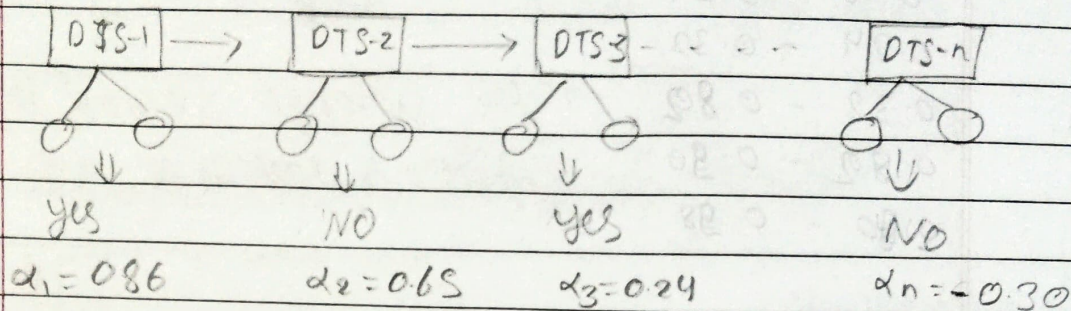
$\rightarrow$ This is send to new stump for calculation.

$\Rightarrow$ And Now, It Iterate over and over.

⑥ Final Prediction $\Rightarrow$

Test data ($\leq 50K, G$)

Steps:



$$f = \alpha_1(m_1) + \alpha_2(m_2) + \alpha_3(m_3) + \dots + \alpha_n(m_n)$$

$$\Rightarrow 0.86(\text{yes}) + 0.65(\text{NO}) + 0.24(\text{yes}) + \dots - 0.30(\text{NO})$$

$$\Rightarrow 1.136(\text{yes}) + 0.350(\text{NO})$$

$\left. \begin{array}{l} \rightarrow \text{Performance to say yes} = 1.136 \\ \rightarrow \text{Performance to say no} = 0.350 \end{array} \right\} \Rightarrow \text{O/P} \Rightarrow \underline{\underline{\text{yes}}}$

$\Rightarrow$ So the Performance of yes is Greater than NO that's why the final output is yes for the Given test data point